

Total No. of Questions : 10]

SEAT No. :

P-9344

[Total No. of Pages : 3

[6181]-579

B.E. (Mechanical)

ENERGY ENGINEERING

(2015 Pattern) (Semester - II) (402047)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8, Q.9 or Q.10.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right of each question indicate full marks.
- 4) Assume suitable data wherever necessary and mention the same clearly.
- 5) Use of steam tables, Mollier chart and calculator is allowed.

Q1) a) Explain the concept of Carbon Credits in brief. [5]

- b) A surface condenser receives 20 t/h of dry saturated steam at 40°C. The air leakage to the condenser is estimated to be 0.35 kg per 1,000 kg of steam. The condensate leaves at a temperature of 38°C. Makeup water is supplied at 10°C. The cooling water enters at 32°C and leaves at 38°C. A separate air extraction pump (SJA) is added and from the air cooler section air along with some steam leaves at 27 °C. The pressure in the condenser is assumed to remain constant. Calculate the rate of saving of condensate. [5]

OR

Q2) a) Which factors are considered for selection of coal in thermal power plant? [5]

- b) What is purpose of ESP? Explain working with simple diagram. [5]

Q3) a) Write Note on : Hydrograph and its applications. [5]

- b) Explain in brief disadvantages of Pulverized Coal Fire Boiler. [5]

OR

Q4) a) Explain the GCR by using following points [5]

- i) Diagram
 - ii) Construction, Working
- b) What are sources of air leakages in condenser? How it affects performance of condenser? [5]

P.T.O.

- Q5) a)** A gas turbine plant of 800 kW capacities takes the air at 1.01 bar and 15°C. The pressure ratio of the cycle is 6 and maximum temperature is limited to 700°C. A regenerator of 75% effectiveness is added in the plant to increase the overall efficiency of the plant. The pressure drop in the combustion chamber is 0.15 bars as well as in the regenerator is also 0.15 bars. Assuming the isentropic efficiency of the compressor 80% and of the turbine 85%, determine the plant thermal efficiency. Neglect the mass of the fuel. [10]
- b)** Explain the applications of diesel power plant. What are environmental impacts of diesel power plant? [8]

OR

- Q6) a)** Air is drawn in a gas turbine unit at 15°C and 1.01 bar and pressure ratio is 7:1. The compressor is driven by the H.P. turbine and L.P. turbine drives a separate power shaft. The isentropic efficiencies of compressor and the H.P, L.P. turbines are 0.82, 0.85 and 0.85 respectively. If the maximum cycle temperature is 610°C, Calculate :
i) The net power developed by the unit per kg/s mass flow.
ii) The work ratio.
iii) The thermal efficiency of the unit.
Neglect the mass of fuel. Take adiabatic index for air gas as 1.4, 1.333 respectively. Take specific heat for air, gas as 1.005 KJ/Kg-K, 1.15 KJ/Kg-K respectively. [10]
- b)** Explain General Layout of Diesel Power Plant indicating different systems. [8]

- Q7) a)** Explain the Binary fluid geothermal power plant with simple diagram. Explain advantages & limitation of the same. [8]
- b)** Explain the Solar Pond based thermal power component arrangement diagram. What are limitation of the same. [8]

OR

- Q8) a)** Explain the working of Close cycle MHD generator with simple figure. Why seed material is injected in MHD generator? [8]
- b)** Write Note on : [8]
i) Fuel Cell
ii) Wind turbine operating characteristics

Q9) a) Write note on following : [8]

- i) Power Transformer
- ii) Relays and its requirements

b) The maximum (peak) load on a thermal power plant capacity of 60 MW is 50 MW at an annual load factor of 50%. The loads having maximum demands of 25 MW, 20 MW, 8 MW and 5 MW are connected to the power station. Determine: [8]

- i) Average load on power station
- ii) Energy generated per year
- iii) Demand factor
- iv) Diversity factor

OR

Q10)a) Write note on following : [8]

- i) Methods of Earthing
- ii) Generator cooling by hydrogen

b) A new factory having a maximum demand of 700 kW and a load factor of 25% is comparing two power supply agencies. Compare both based on cost of energy [8]

- i) Public supply tariff is Rs. 40 per kW of maximum demand plus 2 paise per kWh

Capital cost = Rs. 70,000

Interest and depreciation = 10%

- ii) Private oil engine generating station.

Capital Cost = Rs. 250,000

Fuel consumption = 0.3 kg per kWh

Cost of fuel = Rs. 70 per tonne

Wages = 0.4 paise per kWh

Maintenance cost = 0.3 paise per kWh

Interest and depreciation = 15%.

